Sensor Placement Under Traffic Shift

Forecast comparison, risk sensitivity and failure cases

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Independent research software extension and comparative study

https://github.com/Zahir-Hossain/stochastic-sensor-placement-extension

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Abstract

A fixed sensor network must detect future traffic that may differ from historical observations. We extend stochastic barrier coverage with strictly chronological Poisson intensity stacking, a separate pseudo-BMA baseline, and expected-miss, predictive-void and conditional value-at-risk objectives. Across 40 seeds in designed scenarios, intensity stacking reduces expected misses by 37.6% under drift, 36.3% under lane shift and 32.8% under lane emergence, relative to pooled historical planning. An unannounced drift reversal exposes the opposite outcome. An unstructured total-variation ambiguity experiment is a negative result: its penalty changes little across plans and its radius excludes several shifted futures. A post hoc bandwidth probe of the one-day AIS demonstration exposes estimator-resolution sensitivity and leaves history versus forecast unresolved. Independent numerical checks, raw results, source fingerprints and reproducible workflows make both the successful and unsuccessful cases auditable.

1 Question and relationship to the base framework

Kim et al. [2] formulate two-dimensional barrier coverage for stochastic straight-line trajectories, using a line-space representation, a log-Gaussian Cox line process and greedy placement with numerical refinement. Their earlier work [1] establishes the uncertain-intensity void-probability perspective. We ask how a network placed before a future traffic shift performs when it plans on forecasts, and what a declared intensity-ambiguity set can guarantee.

The implementation adds chronological forecast comparison, distinct intensity-stacking and pseudo-BMA baselines, a tail-risk comparison, a finite-grid TV ambiguity experiment and a reproducible evaluation. The representation, LGCP family, submodular approximation principle, stacking, CVaR and robust observation-selection tools are established methods [3, 5, 6, 8]. This report focuses on their integration, evaluation and failure modes for shifted line traffic.

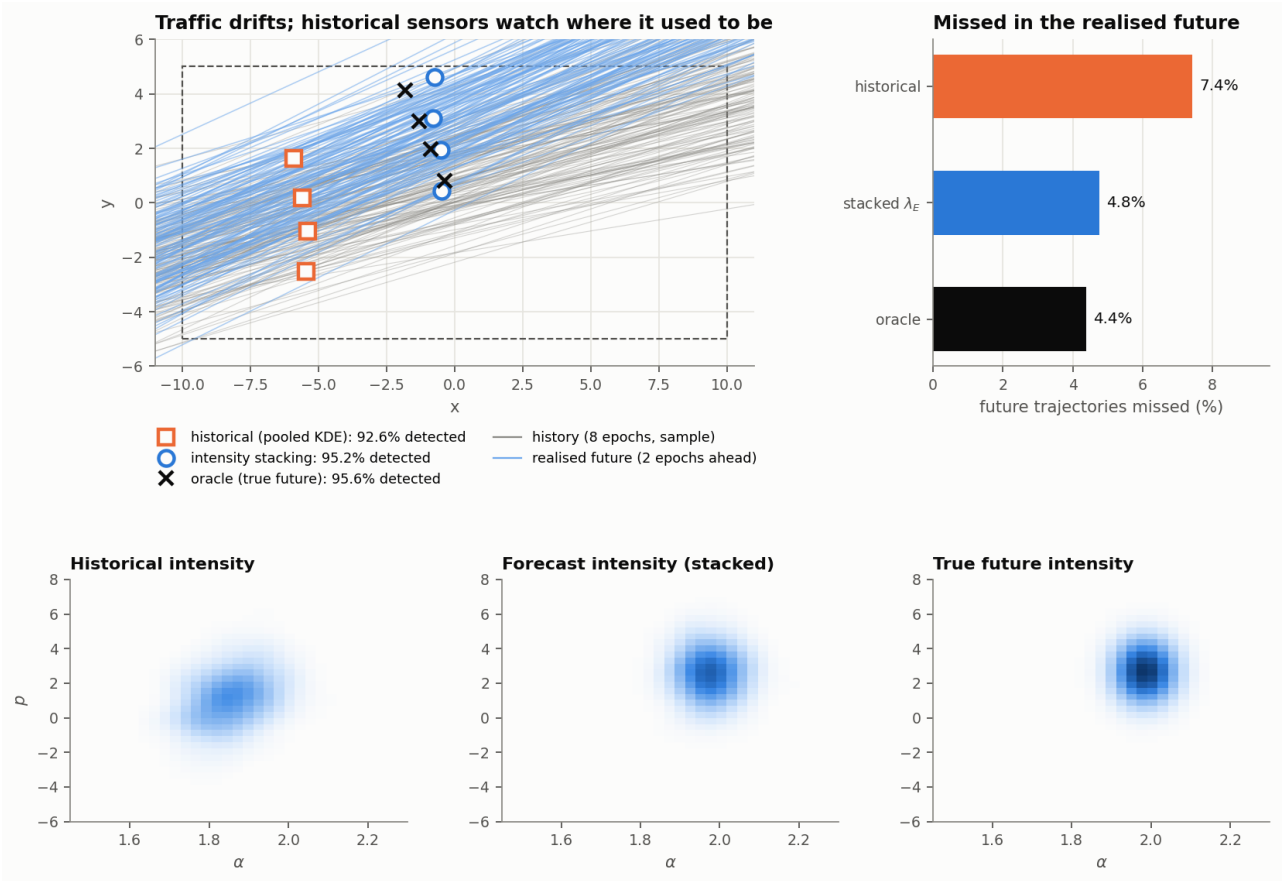


Figure 1: A drifting-lane example. Sensor placements planned on historical and forecast intensities, and a true-future heuristic reference. The lower panels show their corresponding intensities in line space.

2 Model and numerical representation

A line is represented by (α, p) through

$$x\cos\alpha + y\sin\alpha = p, \qquad \alpha \in [0,\pi), \quad p \in \mathbb{R}.$$

For $y = mx + b$, $\alpha = \pi/2 + \arctan m$ and $p = b/\sqrt{1+m^2}$. The identification $(\alpha, p) \sim (\alpha+\pi, -p)$ must be respected at the representation seam. The perpendicular distance from sensor $s = (x_s, y_s)$ to a line is

$$d_s = |x_s\cos\alpha + y_s\sin\alpha - p|.$$

With conditionally independent sensor detections, $q_S(\ell) = \prod_{s\in S}(1 - P_D(d_s(\ell)))$. Independent Poisson thinning gives, for fixed line intensity λ,

$$L(S;\lambda) = \int \lambda q_S, \qquad P(\text{no missed trajectory}) = e^{-L(S;\lambda)}. \tag{1}$$

Expected detection fraction is $1 - L/\int\lambda$; it is not the probability of detecting every trajectory in an epoch.

The grid spans the full angle range and a signed-offset range covering the deployment window. Gaussian kernels are integrated over cells, with seam images. Sensor detection is evaluated at cell

centres and cells are masked using their central line. Thus numerical detection scores are quadrature approximations. Synthetic plans are evaluated on a three-times finer grid along each axis.

Historical intensities are estimated by KDE or a lattice LGCP with Poisson cell counts and precision $Q = \tau(\kappa^2 I + L_g)^2$, where L_g is a graph Laplacian including the seam. The posterior uses a Laplace approximation with plug-in hyperparameters. This is a lattice, SPDE-inspired model [4], not R-INLA or a reproduction of the base paper's inference pipeline [7].

3 Forecasting and chronological combination

Epochs $0, \ldots, T-1$ are observed and the target is $T-1+h$. Forecasts include pooled history, the two most recent epochs, centroid drift, uncertainty-expanded drift and a clipped cellwise log-linear trend. Drift coordinates are lifted continuously across the representation seam. Gaussian shift uncertainty widens a Gaussian kernel by adding covariance; its estimated covariance and small-sample inflation are modelling choices, not a calibrated posterior for every traffic regime.

For held-out target u, training uses only the prefix of length $u-h+1$. Bandwidth is refitted on that prefix. For component intensities $\lambda_{k,u}$, intensity stacking selects simplex weights to maximise

$$\sum_u \left[\sum_{\ell \in D_u} \log\left(\sum_k w_k \lambda_{k,u}(\ell) \right) - \sum_k w_k \int \lambda_{k,u} \right]. \tag{2}$$

The negative objective is convex in w, and is optimised with an analytic gradient and a common numerical log floor. Final component forecasts are then fitted to all available history. A Poisson process at the combined intensity differs from a mixture of full predictive point-process laws. The stacking literature [8] motivates combination but is not invoked to assert an equivalence between those models.

The separate pseudo-BMA baseline uses the softmax of each component's held-out total score. Neither method's weights are asserted to be Bayesian posterior model probabilities. A fixed equal-weight mixture retains the original prototype's idea as another baseline.

4 Objectives and the limits of their guarantees

For predictive intensity Λ, the expected objective needs only its mean: $\mathbb{E}L(S;\Lambda) = L(S;\mathbb{E}\Lambda)$. This identity provides a sufficient posterior-mean interpretation when a posterior is specified; it does not imply that posterior weights are necessary for a particular optimal plan.

We compare expected misses, predictive void loss $-\log \mathbb{E}e^{-L}$ and upper-tail CVaR. Predictive void loss is a soft minimum of sampled losses, emphasising favourable futures; it is not a worst-case penalty. For stacking, samples select one forecast component at a time using its fitted weight. This defines a probability law, but neither a posterior interpretation of the weights nor calibration is established. Favourable component draws can dominate the soft minimum. For equally weighted sampled losses l_i, CVaR at level β is

$$\min_\eta \left[\eta + \frac{1}{n(1-\beta)} \sum_i (l_i - \eta)_+ \right].$$

The implementation includes fractional mass at the tail boundary. For losses $0, 1, \ldots, 31$ at level 0.9, the answer is 29.875; rounding up to four equally weighted observations would instead give 29.5.

The predictive intensity samples for void and CVaR have not been validated for calibration. These comparisons are sensitivity analyses of the supplied law, not decisions under established uncertainty. Both stacking and pseudo-BMA use component selection. The weights are valid sampling probabilities, but neither their posterior interpretation nor uncertainty calibration is established. The fixed mixture carries no samples and uses only its mean field. Weighted-sum sampling would be a different modelling proposal; matching the mean would not by itself validate that proposal.

For expected detected mass, the standard greedy guarantee is $1 - 1/e$ relative to the best subset of the finite candidate set [5]. It does not carry over to missed mass, continuous placement or nonlinear risk objectives. Continuous refinement and nonlinear-objective greedy search remain heuristics.

4.1 A sensitivity diagnostic

Let $\hat{S}$ be a plan made using $\hat{\lambda}$ and S_{ref} the same heuristic given true future intensity λ. Define $R = L(\hat{S}; \lambda) - L(S_{\mathrm{ref}}; \lambda)$ and $\delta = L(\hat{S}; \hat{\lambda}) - L(S_{\mathrm{ref}}; \hat{\lambda})$. Then

$$R \le \max(\delta, 0) + \int |\lambda - \hat{\lambda}| \max(1 - q_{\hat{S}}, 1 - q_{S_{\mathrm{ref}}}) \le \max(\delta, 0) + \|\lambda - \hat{\lambda}\|_1. \tag{3}$$

This follows by linearity and $|a - b| \le \max(1 - a, 1 - b) \le 1$ for $a, b \in [0, 1]$. It is an elementary sensitivity inequality. The reference is not a certified optimum, δ is a measured comparison term and neither truth nor the reference is observable at deployment. Simulation checks verify numerical consistency, not an independent generalisation guarantee. Equal-total Poisson score and Pinsker's inequality motivate log-score validation, but do not guarantee lower regret on every future.

4.2 Finite-grid TV ambiguity

For nominal active-cell masses $m \ge 0$, total $N = \sum_i m_i$ and $\epsilon \in [0, 1]$, define

$$\mathcal{M}_\epsilon(m) = \{z \ge 0 : \mathbf{1}^T z = N,\ \tfrac{1}{2}\|z - m\|_1 \le \epsilon N\}, \qquad \rho_\epsilon(S; m) = \max_{z \in \mathcal{M}_\epsilon(m)} z^T q_S.$$

Proposition 1 (Exact finite-grid inner adversary)**.** *A maximiser transfers up to ϵN mass from cells in increasing order of miss probability to a maximum-miss cell, stopping once remaining donors have that maximum miss probability. Moreover,*

$$m^T q_S \le \rho_\epsilon(S; m) \le \min\{N q_{\max}, m^T q_S + \epsilon N (q_{\max} - q_{\min})\}.$$

Proof. Every feasible change transfers at most ϵN mass. A unit of recipient mass contributes most at $q_{\max}$. Given that recipient, removing mass from a higher-loss donor while a lower-loss donor has mass cannot improve the objective, so an exchange argument orders donors. No positive gain remains after nonmaximum donors are exhausted. □

If normalised grid truth p satisfies $\frac{1}{2}\|p - m/N\|_1 \le \epsilon$, then its expected detection is at least $1 - \rho_\epsilon/N$. This certificate is conditional on the declared grid and ambiguity set. The inner solution is checked against a separate LP; the outer sensor problem uses greedy initialisation and one-sensor exchanges, with no global approximation claim. Choosing a radius does not establish that future traffic lies inside the ball.

5 Evaluation and findings

5.1 Main synthetic comparison

Four designed regimes use eight history epochs, four Gaussian sensors, a two-epoch horizon, 40 seeds and known future intensity. Comparisons are paired by seed, with percentile bootstrap intervals. Candidate spacing is one unit and the deployment window is 20×10 units. The configs and raw outputs are the authoritative experimental specification.

Table 1: Gap to the true-intensity heuristic reference in percentage points, lower is better. Intervals are paired-study summaries over 40 seeds. The reference is not a global optimum.

Planner	Stationary	Drift	Lane shift	Emergence
Historical, KDE	0.24 [0.19, 0.31]	2.98 [2.82, 3.17]	1.40 [1.37, 1.43]	1.98 [1.44, 2.52]
Historical, LGCP	0.23 [0.18, 0.29]	3.45 [3.23, 3.69]	1.45 [1.42, 1.48]	2.22 [1.70, 2.73]
Recent epochs only	0.30 [0.25, 0.36]	1.98 [1.75, 2.23]	0.29 [0.24, 0.34]	0.33 [0.20, 0.48]
Drift extrapolation	0.26 [0.21, 0.31]	**0.18 [0.15, 0.21]**	1.53 [1.46, 1.59]	1.87 [1.51, 2.26]
Drift + uncertainty	0.37 [0.31, 0.44]	0.19 [0.16, 0.23]	1.92 [1.85, 1.98]	2.39 [2.06, 2.69]
Log-linear trend	0.71 [0.63, 0.79]	2.22 [1.99, 2.47]	0.30 [0.25, 0.36]	**0.08 [0.02, 0.14]**
λ_E with fixed weights (v0.1 idea)	0.29 [0.23, 0.34]	0.85 [0.81, 0.90]	1.47 [1.42, 1.51]	2.43 [2.05, 2.79]
Pseudo-BMA (score-weighted baseline)	0.22 [0.17, 0.28]	0.19 [0.16, 0.23]	0.30 [0.25, 0.36]	0.44 [0.23, 0.70]
Stacked λ_E (ours)	0.23 [0.18, 0.28]	0.20 [0.16, 0.24]	**0.23 [0.19, 0.29]**	0.29 [0.15, 0.46]
Stacked λ_E, expected void objective	0.25 [0.19, 0.31]	3.67 [2.90, 4.44]	0.24 [0.19, 0.29]	0.32 [0.22, 0.42]
Stacked λ_E, CVaR objective	0.31 [0.24, 0.38]	0.36 [0.29, 0.45]	0.61 [0.47, 0.77]	1.05 [0.79, 1.32]
Drift + uncertainty, expected void objective	**0.22 [0.19, 0.24]**	0.19 [0.16, 0.23]	1.84 [1.70, 2.01]	1.91 [1.60, 2.22]
Drift + uncertainty, CVaR objective	0.97 [0.84, 1.12]	0.25 [0.21, 0.30]	3.43 [3.21, 3.67]	1.60 [1.37, 1.82]
Log-linear trend, CVaR objective	1.65 [1.42, 1.88]	1.62 [1.40, 1.88]	0.69 [0.56, 0.83]	0.60 [0.42, 0.82]
Evenly spaced fence	12.2	17.1	27.6	25.2
Random	29.1	41.8	28.8	30.6

Intensity stacking changes expected misses relative to pooled history by 0.5% [-0.2, 1.5] in stationary traffic, 37.6% [36.1, 39.1] under drift, 36.3% [34.7, 37.8] under lane shift and 32.8% [24.8, 39.2] under emergence. These are relative reductions, not percentage-point detection gains. The successful regimes closely match candidates: constant-velocity drift matches the drift model, logit-linear lane shares resemble cellwise log-linear trend, and stationarity matches pooled history. These gains show selection of a well-specified candidate, not generalisation to shifts outside every candidate family. The reversal is outside those families: history detects 94.03% versus stacking's 89.53%. This is an experimental-design limitation affecting the headline results, not an implementation bug. The generators do not establish how frequently these regimes arise operationally.

Expected-void planning on the stacked forecast is a separate failure under drift: its reference gap is 3.67 [2.90, 4.44] percentage points, with 29 of 40 seeds above one point. Its detection difference from history is -0.69 [-1.46, 0.11] points. The interval includes zero, so this is a lower mean in these runs, not an established loss across seeds. The gap's correlation with non-drift component weight is 0.70. The component sampler and the soft minimum's preference for favourable draws are relevant mechanisms; correlation alone does not establish causation.

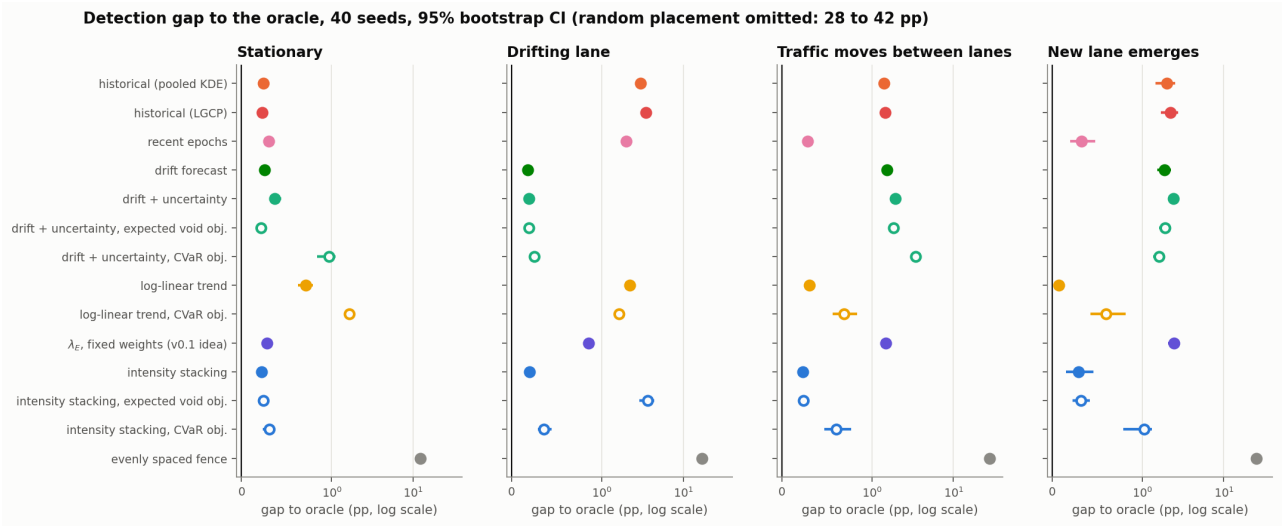


Figure 2: Main synthetic comparison, evaluated on the true future. The oracle label denotes the true-intensity heuristic reference.

5.2 Stress tests and the cost of an unstructured ambiguity set

The robustness study uses 20 paired seeds in six scenarios, common finite-grid candidates, and radii 0, 0.05 and 0.15. All planners use one-sensor exchange refinement. Unlike the main experiment, no continuous refinement is used, so absolute rates should not be compared across the two studies as if only the risk objective had changed.

When drift reverses just after training, historical placement detects 94.03% of future trajectories while intensity stacking detects 89.53%. The turn is not revealed to forecasters. This is an explicit counterexample to a universal benefit from extrapolation.

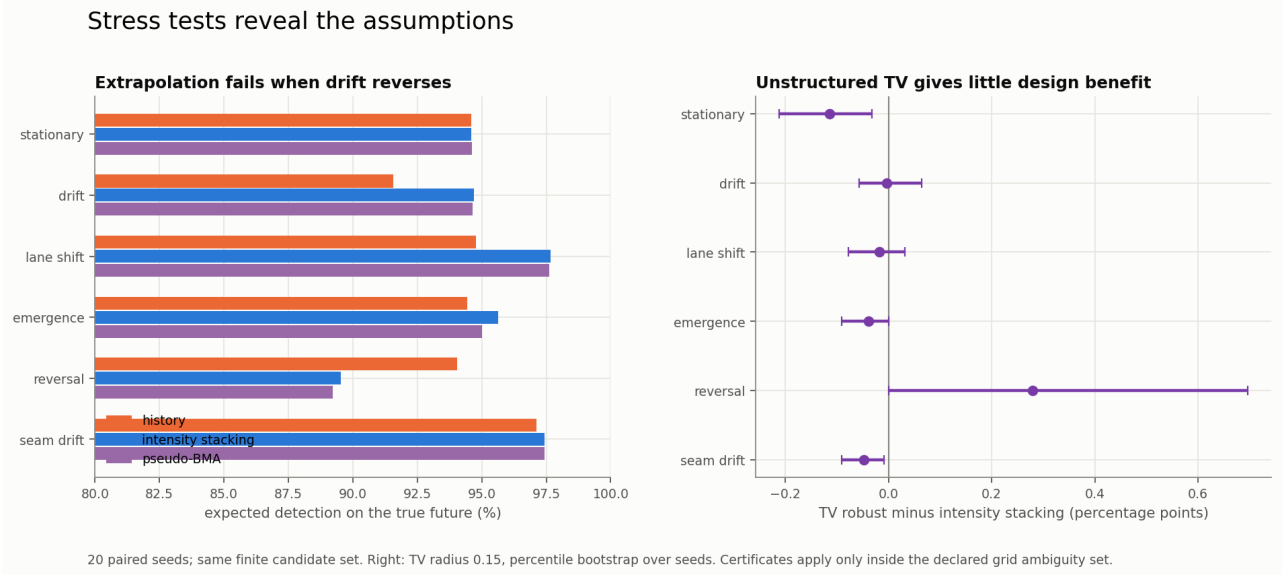


Figure 3: Stress tests expose both forecast failure and the limited design leverage of unrestricted TV ambiguity. Right: change in future detection, TV radius 0.15, with paired-seed bootstrap intervals.

Table 2: TV robust placement minus intensity stacking, radius 0.15. Future change uses finer-grid truth; worst-case change uses the planning grid. The final column is the fraction of simulated truths inside the stated ambiguity ball, not an estimated operational coverage guarantee.

Scenario	Future change (pp)	Worst-case change (pp)	Inside ball
Stationary	-0.114	+0.041	95%
Drift	-0.004	+0.052	95%
Lane shift	-0.018	+0.016	0%
Emergence	-0.039	+0.003	0%
Reversal	+0.279	+0.101	0%
Seam drift	-0.047	+0.006	95%

The TV penalty can change little across plausible plans: many grids have almost-uncovered recipient cells and well-covered donor mass, making the adversarial surcharge close to ϵN across different placements. The experiment therefore does not establish a substantial TV design gain. Across all saved strategies, nominal minus worst-case detection spans 4.93 to 5.00 points at radius 0.05 and 14.73 to 14.99 points at 0.15. At radius 0.05, no scenario-strategy group has truth coverage above 0%. At 0.15, stacked-forecast coverage is 95% for stationary, 95% for drift and 95% for seam traffic; it is 0% for lane shift, 0% for emergence and 0% for reversal. These are grid-based simulated coverage fractions, not a calibration guarantee. A few small negative worst-case differences also expose the limits of the heuristic outer search. Geometry-constrained ambiguity and calibrated radii are useful next research questions, not completed results here. The maximum absolute coarse/fine detection difference in this study is 0.18 percentage points, reinforcing why certificate and fine-grid score must be kept distinct.

5.3 AIS demonstration

A checksummed public sample contains 84,702 AIS reports near Gothenburg on 5 July 2017, yielding 265 straight pieces after filtering. Reports are split at time gaps, window exits, low-speed reports and three-hour epoch boundaries before line fitting. Four rolling origins plan on earlier epochs and score the next epoch's observed pieces.

Historical KDE has mean expected detection 76.6% and intensity stacking 77.7%. At the default bandwidth, historical LGCP leads the KDE planners by 7.5 to 8.6 percentage points. A post hoc multiplier of 0.25 gives historical KDE 84.8%, stacking 85.0% and LGCP 85.2%; recent epochs remain 2.4 points behind LGCP. Historical and stacked KDE thus approach LGCP at finer resolution, but this does not establish equivalence. This probe was computed after viewing the test epochs and is sensitivity evidence only. No scale is adopted. It does not settle history versus forecast. The one-day sample is too small for a broad real-world claim. Table 3 reports every origin and paired differences from historical KDE rather than bootstrap intervals. Origins overlap in training history and do not support population-level significance claims. Metrics use modelled detection per straight piece, not actual sensor measurements or necessarily distinct ships.

Table 3: AIS expected detection (%) at each origin, with the paired change from historical KDE in brackets (percentage points). These are individual test-origin results, not confidence intervals.

Planner	Origin 4	Origin 5	Origin 6	Origin 7
Historical KDE	81.3 [+0.0]	79.2 [+0.0]	69.6 [+0.0]	76.3 [+0.0]
Recent epochs	79.0 [-2.3]	78.0 [-1.2]	79.9 [+10.3]	73.8 [-2.5]
Historical LGCP	87.6 [+6.3]	85.1 [+5.9]	82.8 [+13.2]	85.3 [+9.0]
Intensity stacking	79.0 [-2.3]	78.5 [-0.8]	78.4 [+8.8]	74.9 [-1.4]
Pseudo-BMA	79.1 [-2.2]	79.2 [-0.0]	69.6 [+0.0]	76.3 [-0.0]
Hindsight heuristic	84.6 [+3.3]	84.6 [+5.4]	87.3 [+17.7]	83.4 [+7.1]

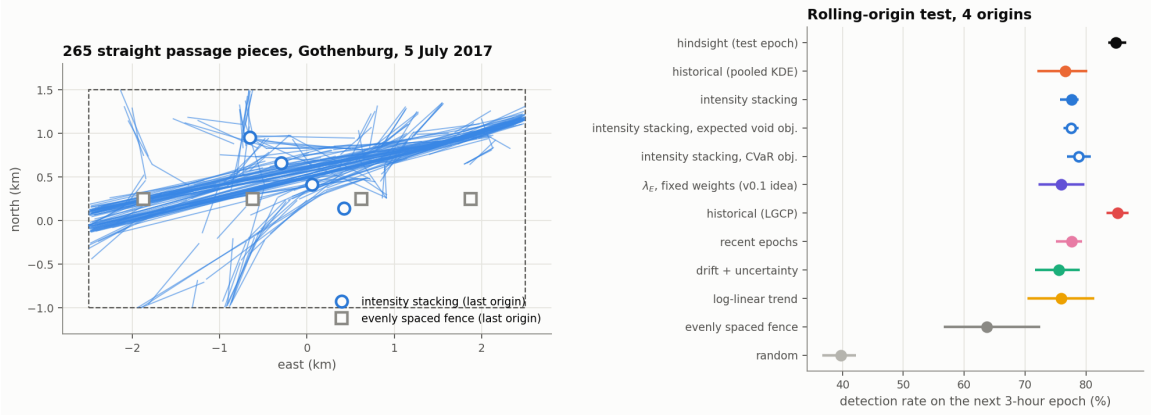


Figure 4: AIS location and descriptive mean summaries. The bars in this retained visualisation resample the same overlapping origins and are not population-level confidence statements; use Table 3 for the individual results. Hindsight fits the test epoch but uses a local heuristic; it is not a strict upper bound.

6 Verification, reproducibility and remaining limits

Tests compare geometry with independent formulas, analytic gradients with finite differences, Poisson thinning with simulated detections, greedy selection with enumeration, fractional CVaR with an epigraph LP and the TV adversary with a separate LP. Regression fixtures enforce chronological bandwidth fitting, seam behaviour and AIS epoch boundaries. A parity test compares the Python legacy port with the unchanged MATLAB script through GNU Octave when that dependency is available.

Each result set records a source SHA-256, configuration, dependency versions and Git revision. The artifact verifier recomputes synthetic summaries, checks run uniqueness and tests every in-scope TV certificate. A Python 3.12 dependency snapshot, VS Code workspace, task definitions, learning notebook and reproduction command are supplied. Hosted CI status must be distinguished from locally completed checks; the validation record in the repository states exactly what ran.

The model assumes straight trajectories, conditionally independent isotropic sensors and distance-only detection. It omits propagation, bathymetry, installation constraints, correlated detection and physical calibration. Synthetic traffic is designed and AIS evidence is limited.

A useful next evaluation would hold out multiple days and sites, include known route changes, calibrate uncertainty before scoring and compare strong domain-specific placement methods. The present result is an auditable study of when forecast-aware placement helps or fails, and why unstructured TV ambiguity adds little design benefit in these examples. Operational deployment remains outside the evidence.

References

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A Preregistered extension evidence

The extension protocol and candidate laws were fixed before these outcomes. All original study summaries are unchanged relative to the corrected pre-extension checkpoint. New methods remain experimental; no sampler or AIS policy replaces a default.

Shifts outside the candidate families (40 paired seeds)

Regime	Stacked minus history, pp [95% CI]	Positive seeds
accelerating	+10.74 [+10.25, +11.27]	40/40
lane_split	+1.00 [+0.63, +1.42]	36/40
random_switch	+2.57 [+1.80, +3.42]	32/40
random_walk	+1.14 [+0.36, +2.07]	26/40
rotating_quadratic	+3.77 [+3.47, +4.10]	40/40

Intervals resample paired seeds. These are specified synthetic generators, not operational traffic.

Sampler diagnostics on the same expected-loss plans

Regime	Sampler	Rank	Coverage	Mismatch
accelerating	component	0.714	62.5%	0.038
accelerating	residual_candidate	0.654	97.5%	0.126
accelerating	weighted_candidate	0.757	42.5%	0.041
lane_split	component	1.000	0.0%	0.015
lane_split	residual_candidate	1.000	0.0%	0.041
lane_split	weighted_candidate	1.000	0.0%	0.008
random_switch	component	0.720	50.0%	0.046
random_switch	residual_candidate	0.770	42.5%	0.072
random_switch	weighted_candidate	0.701	22.5%	0.031
random_walk	component	0.312	10.0%	0.020
random_walk	residual_candidate	0.380	42.5%	0.055
random_walk	weighted_candidate	0.293	5.0%	0.014
rotating_quadratic	component	0.916	32.5%	0.039
rotating_quadratic	residual_candidate	0.662	100.0%	0.190
rotating_quadratic	weighted_candidate	0.922	32.5%	0.043

The gate requires every regime to have at least 80% coverage, mean rank in $[.35, .65]$ and mean relative sample-mean mismatch at most .10. Eligibility for further study: weighted_candidate fails; residual_candidate fails. No sampler is adopted. These are latent-loss ensemble diagnostics, not realized-count PIT or validation of the full point-process law.

Training-only AIS bandwidth selection

Origin	Method	Scale	Detection	Paired change, pp
4	pooled	0.25	87.35%	+6.04
4	recent	0.25	88.22%	+6.91
4	stacked	0.25	87.53%	+6.23
5	pooled	0.25	84.53%	+5.30
5	recent	0.25	83.65%	+4.42
5	stacked	0.25	84.36%	+5.13
6	pooled	0.25	75.02%	+5.44
6	recent	0.25	80.54%	+10.95
6	stacked	0.25	81.07%	+11.48
7	pooled	0.25	92.24%	+15.96
7	recent	0.25	78.80%	+2.53
7	stacked	0.25	77.23%	+0.96

Changes use the default historical KDE reference. Selection uses targets strictly before each test origin. The previously viewed one-day sample is reused: this is exploratory reanalysis, with no population significance claim.

AIS policy ablations (mean over four origins)

Method	Policy	Detection	Paired change, pp
pseudo_bma	both	76.60%	+0.00
pseudo_bma	minimum_origins	76.60%	-0.00
pseudo_bma	original	76.04%	-0.56
pseudo_bma	relative_floor	76.04%	-0.56
stacked	both	78.44%	+1.84
stacked	minimum_origins	78.44%	+1.84
stacked	original	77.66%	+1.06
stacked	relative_floor	77.66%	+1.06

Origin-level weights, solver status and all individual detections are retained in the raw CSV and the generated AIS policy table. Minimum-origin and relative-floor rules are ablations only; the original policy remains the default.

Geometry-aware ambiguity (20 paired seeds)

Regime	Ball	Future gain, pp [95% CI]	Inside	Changed	WC gain
drift	tv	-0.476 [-0.858, -0.095]	100%	5/20	+0.045
drift	w1	-10.378 [-12.405, -7.865]	100%	20/20	+6.548
emergence	tv	+0.000 [+0.000, +0.000]	65%	0/20	+0.000
emergence	w1	-5.274 [-5.738, -4.757]	50%	20/20	+3.763
lane_shift	tv	-1.085 [-1.640, -0.586]	100%	10/20	+0.134
lane_shift	w1	-18.263 [-20.759, -16.684]	100%	20/20	+10.293
reversal	tv	-0.519 [-1.618, +0.428]	20%	3/20	+0.041
reversal	w1	-16.070 [-16.070, -16.070]	30%	20/20	+10.735
seam_drift	tv	-2.816 [-3.840, -1.792]	100%	11/20	+0.129
seam_drift	w1	-2.114 [-3.423, -1.052]	100%	8/20	+0.121
stationary	tv	+0.000 [+0.000, +0.000]	100%	0/20	+0.000
stationary	w1	-0.835 [-0.982, -0.688]	95%	17/20	+0.786

The transport plan lowers mean future detection in 6/6 tested regimes despite improving the objective it optimizes. This is a negative empirical result; geometric structure alone did not improve these placements.

The geometry-aware finite-grid transport dual was checked against an independent primal linear program. Every reported certificate is conditional on normalized grid truth lying in its ball. Three chronological training errors do not establish nominal coverage. Radii have different units across TV and transport. Search is discrete and heuristic; no global placement or operational guarantee is claimed.

Multi-week AIS (C7) The proposed July 2020 routing-change study remains blocked by archive access. No raw archive, checksum or multi-week result is reported. Settings and the data dependency are recorded in the repository.